

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10: SUB-STRAND 2.8 VECTORS, FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: Why do two matatu journeys of exactly the same distance from Nairobi's CBD end up at completely different destinations, and what mathematical tool do we need to describe a journey completely? Answer every section in the space under each prompt. Use what you learned in Lessons 1 to 8: the Kencom matatu map, the Nairobi grid with Kencom Bus Stop as the origin O, and the Nakuru journey map. In the calculation sections, show each step of your working and write the correct units (take 1 grid unit = 1 km). Use the terms scalar, vector, distance and displacement carefully. Where a question asks you to explain, write full sentences and connect your answer back to the two matatus from Kencom.

Part 1: Why is '8 km' not enough?

Two matatu drivers both travel 8 km from Kencom Bus Stop in Nairobi's CBD. One arrives at Westlands (north-west on the map) and the other arrives at South B (south-east on the map). (a) Is '8 km' a scalar or a vector quantity? Explain what it tells you and what it does not tell you. (b) Give one example of a scalar and one example of a vector from everyday Kenyan life, and say why each belongs in its group. (c) Use the words magnitude and direction to explain why the two matatus end up in different places.

(a) '8 km' is a scalar. It gives only the magnitude (how far), so it tells us the size of the journey but not which way the matatu went. (b) Scalar example: the time to cook ugali (20 minutes), or the temperature in Nairobi (22°C). Each is described completely by a size alone. Vector example: the velocity of a bodaboda heading south on Ngong Road, or the displacement of the Likoni Ferry from the mainland to the island. Each needs both a size and a direction to be complete. (c) Both journeys are labelled with the same magnitude, 8 km, but their directions are different: Route A heads north-west towards Westlands and Route B heads south-east towards South B. Starting from the same point, Kencom Bus Stop, a different direction gives a different endpoint, so one matatu reaches Westlands and the other reaches South B. A journey is described completely only by magnitude and direction together, which is a vector.

Part 2: Writing journeys as column vectors

On the simplified Nairobi coordinate grid used in Lessons 5 and 6, Kencom Bus Stop is the origin O. On this

(a) The top number is the horizontal movement (positive means east, negative means west). The bottom number is the vertical movement (positive means north, negative means south). (b) $|w| = \sqrt{((-6)^2 + 5^2)} = \sqrt{(36 + 25)} = \sqrt{61} \approx 7.8$ km. $|s| = \sqrt{(4^2 + (-6)^2)} = \sqrt{(16 + 36)} = \sqrt{52} \approx 7.2$ km. (c) Both magnitudes are close to the 8 km on the map but are not exactly 8 km, because the grid coordinates are approximate readings from a simplified map. The point is that both journeys have a size of roughly 8 km, so the

grid, one Westlands-route journey is the column vector $w = (-6; 5)$ and one South B-route journey is the column vector $s = (4; -6)$. (a) Explain what the top number and the bottom number in a column vector tell you. (b) Calculate the magnitude of w and the magnitude of s, correct to 1 decimal place. (c) Compare your two magnitudes with the '8 km' labels on the map. (d) Explain why the two column vectors are different even though their magnitudes are similar.	scalar distance alone cannot tell them apart. (d) The magnitude is a scalar, so it keeps only the size and discards the direction. Each component of w has the opposite sign to the matching component of s, so they point in roughly opposite directions and are not equivalent vectors. Their endpoints are therefore far apart, which is why the matatus arrive at different destinations. With east and north positive, w points north-west towards Westlands and s points south-east towards South B, matching the map.
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Part 3: Position vectors, displacement and midpoint

On the same grid, KICC is at point A and Uhuru Park is at point B. The position vectors are $OA = (2; 3)$ and $OB = (-1; 2)$. (a) Find the displacement vector AB. (b) Find the magnitude of AB, correct to 1 decimal place. (c) A matatu travels from A to B. Find the coordinates of the midpoint of this journey. (d) A translation moves every point of an object by the same vector. State the translation vector that moves point A to point B and explain how you found it.	(a) $AB = OB - OA = (-1 - 2; 2 - 3) = (-3; -1)$. (b) $AB = \sqrt{(-3)^2 + (-1)^2} = \sqrt{9 + 1} = \sqrt{10} \approx 3.2$ km. (c) Midpoint $M = ((2 + (-1))/2; (3 + 2)/2) = (0.5; 2.5)$. (d) The translation vector that moves A to B is $(-3; -1)$. It is found as image position vector minus object position vector, $OB - OA$, which is the same calculation as the displacement vector AB.
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Part 4: Adding, scaling and reversing journeys

(a) A bodaboda rider goes from O to A, then from A to B, using the position vectors from Part 3. Write the two legs as column vectors, add them to find the resultant, and say what the	(a) The first leg is $OA = (2; 3)$ and the second leg is $AB = (-3; -1)$. The resultant is $(2; 3) + (-3; -1) = (-1; 2)$. This equals OB, the position vector of Uhuru Park, so the resultant is the single displacement from the start O to the final point B. (b) Draw the first vector from O to A. Then draw the second vector starting at the tip of the first, so it goes from A to B. The resultant is the single arrow from the tail of the first vector (O) to the tip of the last vector (B). (c) $3d = (6; 3)$. It has three times the magnitude of d and the same direction. $-d = (-2; -1)$. It has the same magnitude as d but the opposite direction. Both stay on the same line as d. (d) The vector from B to O is $-OB = (1; -2)$. It has the same magnitude as OB but the opposite direction, so adding it to OB gives $(0; 0)$
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resultant represents on the grid. (b) Explain in words how the triangle law (tip-to-tail) would show this resultant on squared paper. (c) The displacement vector $d = (2; 1)$. Write $3d$ and $-d$ as column vectors and describe how each compares with d in magnitude and direction. (d) Write the journey from B back to O as a vector and explain how it is related to OB.	and the journey returns to the start.
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Part 5: Final explanation and the Nakuru journey

Write a full answer to the driving question: Why do two matatu journeys of exactly the same distance from Nairobi's CBD end up at completely different destinations, and what mathematical tool do we need to describe a journey completely? In your answer: (a) use the terms scalar, vector, magnitude and direction. (b) Refer to your column vector results from Part 2. (c) Explain how you would use vectors to describe a three-leg journey in Nakuru from the CBD Terminus to Westside Mall, then the ASK Showground, then the Lake Nakuru Viewpoint. Say how the total distance travelled differs from the displacement from start to finish. (d) Describe how your thinking has changed since Day 1.	The two matatus end up in different places because the distance '8 km' is a scalar. It gives only the magnitude of the journey and says nothing about direction. Both drivers are labelled with the same distance from Kencom, but their directions differ, so their displacement vectors differ. In Part 2, $w = (-6; 5)$ and $s = (4; -6)$ have magnitudes of about 7.8 km and 7.2 km. These are both roughly 8 km on our simplified grid, but not exactly equal to it or to each other, so even the magnitude cannot separate the journeys cleanly; what separates them is that their components point in opposite directions, so the endpoints are Westlands and South B. The tool we need is the vector, which carries both magnitude and direction and can be written as an arrow or as a column vector. For the Nakuru journey, I would write each leg (CBD Terminus to Westside Mall, Westside Mall to the ASK Showground, ASK Showground to the Lake Nakuru Viewpoint) as a column vector. I would add the three column vectors component by component to get the resultant displacement from the start to the finish. I would find its magnitude with Pythagoras' theorem, $\sqrt{x^2 + y^2}$. The total distance travelled is the sum of the lengths of all three legs, but the displacement is the single straight-line vector from start to finish, so the two are generally not equal. The translation vector from the start to the finish is the same as this resultant. On Day 1 I thought that travelling the same distance should bring you to the same place. Now I understand that distance is a scalar, and that direction as well as size is needed to describe a journey completely and to predict where it ends.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
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Scalar and vector concepts	Correctly identifies 8 km as a scalar, gives accurate Kenyan examples of both types, and clearly explains that magnitude and direction together describe a journey.	Identifies scalar and vector correctly with mostly accurate examples, but the explanation of why direction matters is brief.	Confuses scalar and vector, or gives examples without a reason.
Column vector and magnitude calculations	Reads components correctly and applies $\sqrt{x^2 + y^2}$ accurately to get 7.8 km and 7.2 km, with units and an honest comparison with the 8 km labels.	Calculates the magnitudes with a minor slip in rounding or units, and makes a limited comparison with the labels.	Cannot interpret components or uses an incorrect formula for magnitude.
Position vectors, resultant, midpoint and translation	Correctly finds $AB = (-3; -1)$, its magnitude, the midpoint $(0.5; 2.5)$, the resultant $(-1; 2)$, the scaled vectors, and links translation to displacement.	Gets most of the vector operations right with one or two arithmetic or sign errors, and links ideas partially.	Makes frequent errors in subtraction, addition or scalar multiplication and does not link the results.
Final explanation of the driving question	Gives a coherent explanation using the vocabulary, the Part 2 evidence, and the multi-leg Nakuru journey, distinguishes distance from displacement, and reflects on how their thinking changed.	Answers the driving question with correct vocabulary but gives limited evidence or only a weak reflection.	Gives an incomplete or unclear answer that does not use the vector ideas or the evidence.

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